MutAAP-IC

See the impact.

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MOTIVATION

Interpreting the impact of mutations requires integrating multiple analyses

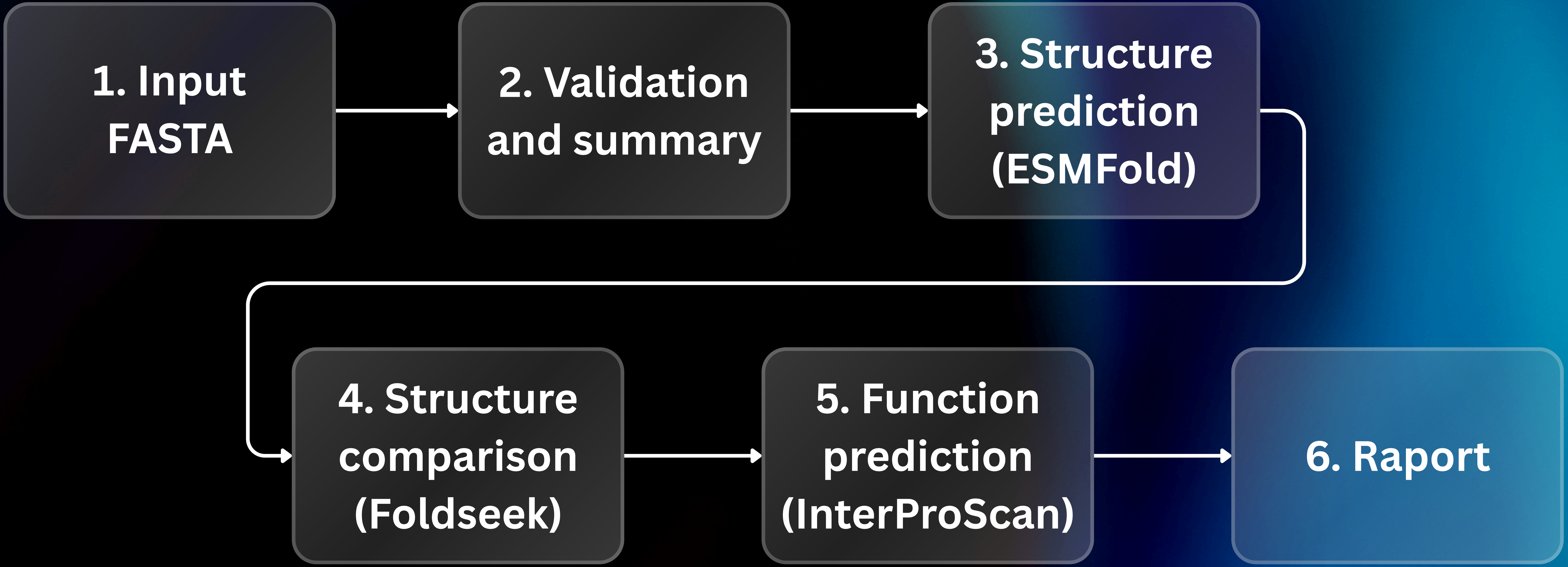
Coding mutations can alter protein structure and function

Researchers need fast, automated evaluation of mutant proteins

PURPOSE OF THE TOOL

An end-to-end tool that takes two nucleotide sequences, verifies ORF integrity, predicts and compares the resulting protein structures, searches for structural homologs, and generates an integrated functional-impact report.

OVERALL WORKFLOW



INPUT REQUIREMENTS, VALIDATION & SUMMARY

- Given FASTA sequence must be an ORF:
    - codon START
    - codon STOP - only at the end
    - divisible by 3
    - will give max 400 aa (ESMfold limit)
- Automatic alignment or changes given by user
- Translation to amino acid sequence
- Mutations in protein and nucleotide sequence
- Summary of the information (.tsv)

STRUCTURE PREDICTION MODULE

- Predicted 3D protein structures using ESMFold via the ESM Atlas API.
- Input: amino acid sequence (original and mutated variants).
- Submitted sequences to the ESMFold server and retrieved predicted structures in PDB format.
- Saved predictions for downstream structural analyses and comparison.
- Implemented automatic error handling:

STRUCTURAL ALIGNMENT & COMPARISON

- Compared predicted structures of wild-type and mutant proteins using TM-align.
- Calculated:
  - TM-score - measure of structural similarity (0–1).
  - RMSD - average deviation between aligned atomic coordinates.
- Generated optimal structural superposition using the rotation matrix from TM-align.
- Extracted residue-level alignments for detailed comparison.

SEARCH FOR STRUCTURAL HOMOLOGS

- Searched predicted protein structures against a Foldseek structural database.
- Performed structure-based similarity search using Foldseek easy-search.
- Retrieved top matching structures based on:
    - TM-score
    - RMSD
    - lDDT
    - E-value
- Ranked hits by structural similarity.
- Returned the top k most similar protein structures.

FUNCTIONAL CONSERVATION ASSESSMENT

- Predicted protein function using the InterProScan web service.
- Annotated protein sequences with three Gene Ontology (GO) terms: Molecular Function (MF), Biological Process (BP), Cellular Component (CC).
- Analyzed and compared both wild-type and mutant protein sequences.
- Classified GO terms as:
  - Conserved - present in both proteins.
  - Lost - present only in the wild type.
  - Gained - present only in the mutant.
- Evaluated potential mutation-induced functional changes.

OUTPUTS

MutAAP-IC Report

Inputs

- Original structure: orig_esmfold_v1.pdb
- Mutant structure: mut_esmfold_v1.pdb

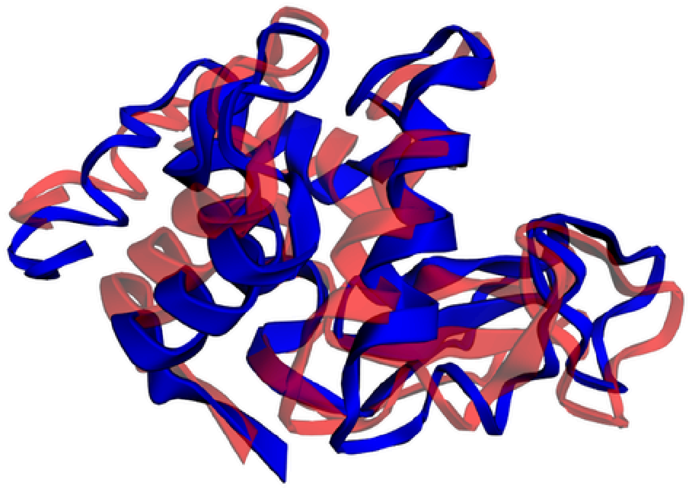
TM-align Comparison

Preliminary interpretation of the mutations

Preliminary interpretation: probably tolerated / lower risk. The structures are highly similar, so the mutation may be less likely to disrupt the protein, but this is not a functional guarantee.

tm_score	0.9956
rmsd	0.280
alignment	KVFGRCELAAAMKRHGLDNYRGYSLGNWVHAAKFESNFNTYKTNRNTDGSTDYGILQINSRWWCNDGRTPGSRNLCNYTRSALLSSDITASVNCAKKIVSDGNGMNAWVAWRNRCKGTDVQAWIRGCRL ::: KVFGRCELAAAMKRHGLDNYRGYSLGNWVCAAKFESNFNTQATNRNTDGSTDYGILQINSRWWCNDGRTPGSRNLCNIPCSALLSSDITASVNCAKKIVSDGNGMNAWVAWRNRCKGTDVQAWIRGCRL

Structure View



Original (blue) Mutant (red)

PDB Foldseek Results

query	target	tm	rmsd	lddt	evalue
mut_esmfold_v1	3zvq-assembly1_A	1.005	0.3596	0.6361	2.737000e-10
mut_esmfold_v1	1lsn-assembly1_A	1.003	0.3587	0.9871	2.468000e-22
mut_esmfold_v1	1fn5-assembly1_A	1.003	0.3642	0.9897	1.958000e-22

AlphaFold Foldseek Results

query	target	tm	rmsd	lddt	evalue
mut_esmfold_v1	AF-Q7LZQ0-F1-model_v6	1.005	0.2833	0.9946	1.368000e-23
mut_esmfold_v1	AF-P00698-F1-model_v6	1.005	0.2864	0.9957	2.052000e-23
mut_esmfold_v1	AF-Q7LZI3-F1-model_v6	1.005	0.2850	0.9924	3.661000e-23

Functional Annotation (InterProScan)

Lost: 0 Gained: 0 Conserved: 1

Molecular Function (MF)

GO Term	Description	WT	Mutant	Status
GO:0003796	lysozyme activity	✓	✓	conserved

Biological Process (BP)

No results.

Cellular Component (CC)

No results.

File: report/changes_report.tsv

```
SUMMARY
original_protein_length:    111
modified_protein_length:    111
frameshift: False

AMINO_ACID_CHANGES
aa_position_in_mut  original_aa modified_aa mutation_type
2   M   L   nonsynonymous

24  A   -   deletion

111 -   *   nonsynonymous


NUCLEOTIDE_CHANGES
nt_position_in_mut  original_nt modified_nt change_type
4   A   T   mismatch
72-72   TGC A-- mismatch
75  T   -   deletion
```

DEPENDENCIES AND USAGE

Requirerments:

- Linux or WSL2
- Python >= 3.10
- Conda (Miniconda/Anaconda)
- Foldseek installed automatically via environment

Step 1 - Clone the repository

- git clone https://github.com/anastazjat08/mutaap-ic.git
- cd mutaap-ic

Step 2 - Create the conda environment

- conda env create -f environment.yaml
- conda activate mutaap_env

Step 3 - Install the MutAAP-IC package

- pip install .
- mutaap *(This installs the CLI command)*

Quick usage example:

```
mutaap orig_fasta_path mut_fasta_path \
 --custom_db custom_structures_folder \
 --exclude_af \
 --top_k 20 \
 --out_dir all_results_path
```

FUTURE FEATURES

- Prediction of longer protein structures
- Searching for ORFs among longer sequences
- Genomic analysis
- RNA and alternative translations
- Protein - protein interactions prediction

SUMMARY

MutAAP-IC is a CLI tool for automated analysis of protein mutations.

Key modules:

- Structure Prediction – predicts 3D structures via ESMFold
- Structural Alignment – compares wild-type vs mutant using TM-align
- Homolog Search – finds similar structures with Foldseek
- Functional Conservation – assesses GO term changes via InterProScan

Outputs structured results and a concise report.

THANK YOU

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